Do ECG Heartbeat Classifiers Generalise to Unseen Patients?

A Controlled Study of a 1D CNN on MIT-BIH

SC4001 Neural Networks and Deep Learning: Group Project Report

Group [number]

Huang Yitian: evaluation and report (M3)

Hou Junning: data and features (M1)

Nguyen Khanh: models and training (M2)

November 2026

1 Introduction

Automatic heartbeat classification from the electrocardiogram (ECG) supports long-term monitoring, where a single day of recording contains around 100,000 beats. The MIT-BIH Arrhythmia Database [1], [2] is the standard benchmark, with beats grouped into the AAMI classes normal (N), supraventricular ectopic (S), ventricular ectopic (V), fusion (F) and unknown (Q) [3].

Reported results depend heavily on how beats are split. If beats from all recordings are pooled and divided at random (the *intra-patient* paradigm), beats from the same person appear in both the training and the test set, so a model can succeed by recognising the person rather than the arrhythmia. Under this paradigm deep models report near-perfect scores, for example 100% S sensitivity at 96.5% positive predictivity [4]. When training and test recordings are disjoint (the *inter-patient* split of de Chazal et al. [5]) the task is much harder: their classifier found 75.9% of S beats at a positive predictivity of only 38.5%, and Sellami and Hwang report 99.48% accuracy intra-patient but 88.34% inter-patient [6]. The same pattern appears across clinical machine learning: record-wise validation often overestimates accuracy compared with subject-wise validation [7], and leakage between training and test data has produced over-optimistic conclusions in many fields [8]. A deployed classifier always meets a new patient, so only a patient-wise evaluation answers the clinically relevant question.

The patient-wise benchmark has a difficulty of its own: abnormal beats are rare and concentrated in a few people. In the standard test set (DS2; 49,661 beats after our preprocessing) S beats make up 3.7% of beats, and one recording, 232, contains 75% of them; recording 213 contains 93% of the test F beats. Predicting N for every beat already scores 89.0% accuracy, with a macro-F1 of only 0.24.

This project asks **how much a 1D convolutional neural network (CNN) loses when it is evaluated on patients it has never seen, and whether inexpensive changes recover the rare abnormal classes.** We train one baseline on a random beat split and on a subject-disjoint patient split, and then test four changes, each aimed at a suspected cause of the gap: class-weighted and focal losses for class imbalance [9], [10], waveform augmentation for differences in recording conditions [11], [12], and RR-interval features, because S beats are mostly early relative to the local rhythm while their shape resembles that of N beats. Our contributions are:

- A controlled measurement of the gap: macro-F1 falls from 0.874 to 0.429, and S and F F1 fall from 0.74 and 0.81 to zero. Our patient split is fully subject-disjoint; records 201 and 202 come from the same person, which the conventional split ignores.
- A comparison of four fixes over three seeds, with per-class sensitivity, positive predictivity and F1, per-patient results and patient-bootstrap intervals. None recovers S detection; class weighting appears to, but only by labelling whole patients as S.
- A failure analysis showing that RR features teach the network to rank S beats correctly *within* a patient (AUROC 0.994 in record 232) even though it never predicts S, and that a decision threshold set on other patients does not fix this.
- A data analysis explaining why timing does not transfer: 62% of training S beats lie inside tachycardia runs, where they are not early, while most test S beats are isolated early beats.

2 Review of Existing Techniques

2.1 Heartbeat classification on MIT-BIH

Class imbalance. ECG datasets are highly imbalanced, with normal beats dominating minority classes, making accuracy an unreliable performance metric [13]. To address this, class-weighted cross-entropy increases the penalty for misclassifying rare classes, while focal loss reduces the contribution of easily classified examples [10]. Previous ECG studies have also addressed imbalance using batch-weighted loss [6] and data augmentation [14]. In this study, we compare inverse-frequency class weighting (E2) and focal loss (E3) against the baseline CNN.

2.2 Timing features and data augmentation

Heartbeat classification can combine waveform morphology with temporal context. de Chazal et al. [5] used morphology and heartbeat-interval features with separate development and evaluation record sets. This motivates comparing an isolated-waveform CNN with one that also consumes RR features. Our four-class neural model and four-feature timing vector differ from their classifier, so we do not claim to reproduce its reported performance.

Augmentation offers a complementary way to test robustness to recording variation. Nonaka and Seita [15] studied augmentation for single-lead atrial-fibrillation classification, a different task from our heartbeat classification. Raghu et al. [16] found that augmentation effects depend on the ECG prediction task and proposed the learned TaskAug policy. We instead fix gain, noise, baseline and alignment perturbations, keeping architecture and evaluation patients unchanged. These studies motivate testing augmentation; they do not establish that

our transforms improve unseen-patient performance. We test these questions in E4 and E5.

2.3 Evaluation protocols and patient-level uncertainty

An evaluation is only meaningful if the relationship between training and test data matches that between training data and future use [7]. Even the conventional inter-patient split is not fully subject-disjoint: records 201 (DS1) and 202 (DS2) come from the same person, and PhysioNet advises keeping them in one partition for subject-independent work [17]. How results are summarised also matters. ANSI/AAMI EC57 emphasises record-by-record reporting [3], yet headline inter-patient results are single figures per class, pooled over DS2. Such pooled figures are dominated by a few recordings, and beats from one patient are not independent, so intervals that treat beats as independent samples are too narrow. Resampling whole clusters, here patients, is the standard bootstrap for clustered data [18], [19], [20]. We therefore report per-record results, results without the dominant records, and patient-bootstrap intervals alongside the pooled metrics.

3 Data and Methods

3.1 Data and preprocessing

We used the MIT-BIH Arrhythmia Database v1.0.0 [17], excluding paced records 102, 104, 107 and 217 and retaining 44 recordings. Original waveform, header and expert annotation files were verified against the publisher's SHA-256 checksums. Beat symbols were grouped into four superclasses: N={N,L,R,e,j}, S={A,a,J,S}, V={V,E} and F={F}, encoded as 0–3. N therefore includes non-ectopic types beyond normal sinus beats. Q-class targets and non-beat annotations were excluded.

We selected MLII by channel name, including the second channel in record 114. Each reference annotation supplied a 256-sample window spanning $[-128, +128)$ samples at the original 360 Hz, approximately 0.711 s. We applied no resampling or bandpass filter. Excluding 15 Q targets, 40 boundary-crossing target windows and 48 further targets without previous/next RR context left 100,630 labelled beats; 2,991 non-target annotations were ignored. All experiments use this retained pool. For each protocol, a scalar waveform mean and population SD were fitted on training beats only and frozen for validation/test. RR features were standardised column-wise in the same way. Annotated centres make this a beat-classification experiment conditional on known beat locations, rather than a complete detection system.

3.2 Partitions

E0 uses a stratified 80/20 development/test split, with 20% of development reserved for validation: approximately 64/16/20 overall. The data seed is fixed at 42 across all model seeds. Subjects and neighbouring-beat context may overlap across these random partitions.

E1–E5 use `patient_strict`, based on the conventional DS1/DS2 record lists [5]. DS1 records 108, 118, 203 and 223 form validation; DS2 is reserved for final evaluation. Since records 201 and 202 belong to the same person [17], we remove record 201 from training. The resulting 17 training, four validation and 22 test subjects are disjoint. This is an explicit modification of the conventional protocol, which remains available as a reference. Table 1 gives the actual retained counts. All model seeds share these partitions and their training-fitted normalisation; protocol prefixes are never mixed. Changing split also changes cohort composition and training size, so the E0/E1 gap cannot isolate patient overlap as its sole cause.

Table 1: Retained beats by partition and class. Random is used by E0; strict patient is used by E1–E5.

Protocol	Partition	Records	Beats	N	S	V	F
Random	Train	44	64,404	57,627	1,778	4,485	514
Random	Validation	44	16,101	14,407	445	1,121	128
Random	Test	44	20,125	18,008	556	1,401	160
Strict patient	Train	17	39,394	35,719	640	2,640	395
Strict patient	Validation	4	9,614	8,472	175	950	17
Strict patient	Test	22	49,661	44,218	1,836	3,219	388

3.3 RR features and augmentation

E5 adds four RR features: previous RR, next RR, the mean of up to ten preceding RR intervals including the current previous interval, and the previous RR divided by that mean, termed the *prematurity index*. The first three are measured in seconds; the index is dimensionless, with values below one indicating an early beat relative to the local rhythm. Features are computed from consecutive beat annotations within each recording, retaining Q beats as timing context but excluding rhythm markers. Intervals never cross record boundaries and class identities are not feature inputs. All four columns, including the index, are standardised using training

statistics. Next-RR requires future context, so E5 is offline classification.

For E4, each training waveform independently receives gain scaling in $[0.9, 1.1]$, Gaussian noise with SD 0.02, sinusoidal baseline wander with amplitude 0.05 and frequency in $[0.1, 0.5]$ Hz with random phase, and an integer shift in $[-8, 8]$ samples, each with probability 0.5. Shifts use edge padding. Additive magnitudes are in training-standardised units. Transform randomness is determined by model seed, epoch and row index. Labels and RR features stay fixed because these perturbations model measurement variation. Validation/test waveforms are never augmented; E0–E4 do not consume RR features.

3.4 Model and training

Models E0–E5 were trained using a maximum of 30 epochs, with early stopping patience of 4, based on validation macro-F1. Each experiment was repeated with three random seeds (42, 43, 44). NVIDIA GPU (CUDA) acceleration is recommended to reduce training time. The validation set contained only 175 supraventricular (S) and 17 fusion (F) beats, with the S beats originating from two records. This limited minority-class representation makes validation macro-F1 sensitive to individual predictions and patient-specific characteristics. Although, early stopping may have selected checkpoints before the models learned minority-class patterns, potentially affecting generalisation.

To reproduce the training:

```
python -m src.train --experiments all --seeds 42 43 44 --epochs 30 --patience 4 --device cuda
```

3.5 Evaluation

Abnormal beats are concentrated in a few records (Fig. 1): record 209 holds 60% of the training S beats and record 208 holds 94% of the training F beats, while in the test set record 232 holds 75% of the S beats and record 213 holds 93% of the F beats. Pooled test scores therefore mostly describe a few patients, and our evaluation is designed around this.

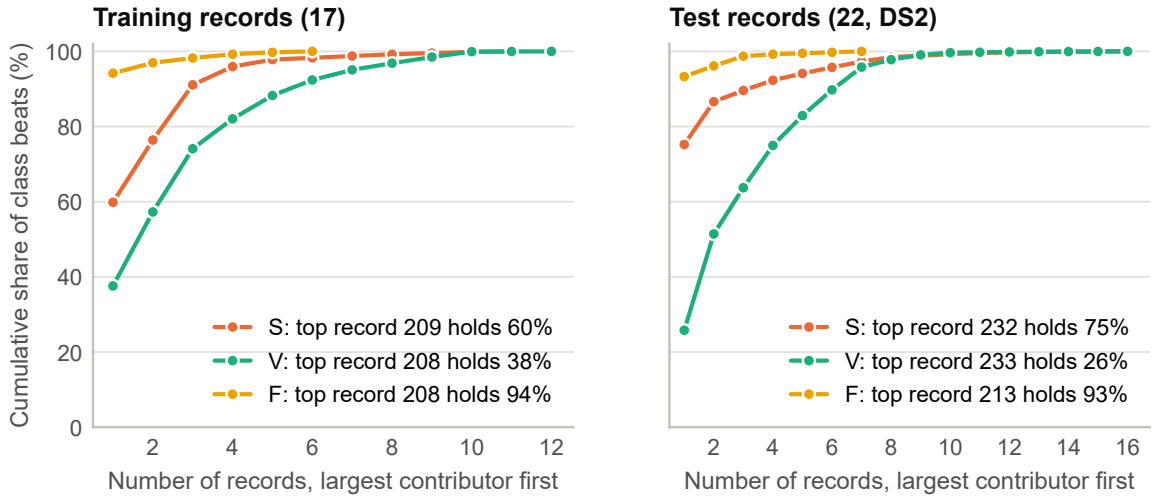


Figure 1: Concentration of abnormal beats: cumulative share of each class's beats, records sorted by contribution. Left: the 17 training records; right: the 22 test records.

We report per-class sensitivity (Se, recall), positive predictivity (+P, precision) and F1, and the macro-F1 over the four classes, as mean ± SD over the three seeds. Test predictions were exported once from the frozen checkpoints. Before any test beat was read, the processed arrays passed the data audit and every checkpoint reproduced its stored validation macro-F1 on our copy of the data (largest difference 3×10^{-4}, from GPU/CPU rounding). Because two records dominate the test S and F beats, we also report S without record 232 and F without record 213 as sensitivity analyses. To show how much a result depends on which patients are tested, we resampled the 22 test records with replacement 2,000 times, using the same draws for every model, and report 95% percentile intervals for each metric and for its paired difference from E1.

4 Experiments and Results

4.1 Experiments and hypotheses

E0 (random split) and E1 (patient split) use the baseline CNN with cross-entropy. E2–E5 each change one element relative to E1: class-weighted loss (E2), focal loss (E3), augmentation (E4) and RR features (E5). Before any training run, we wrote down five hypotheses: the gap is large and concentrated in S and F (H1);

weighted and focal losses raise rare-class sensitivity at the cost of +P (H2); augmentation helps little (H3); RR features give the largest S gain (H4); and no conclusion about F will be robust (H5). A sixth hypothesis, on autoencoder pretraining (E6), was not tested: M1 pretrained the encoders, but no supervised E6 classifier was trained (Appendix B). Two analyses in Section 4.4 were added after the first test evaluation and are marked *post-hoc*; no model or threshold was selected on test data.

4.2 The generalisation gap (H1: supported)

Table 2: Main results on the full test set (mean ± SD over three seeds). n/a: the class was never predicted. *Only one seed predicted any S beat.

Exp	Model (test set)	Macro-F1	S Se	S +P	S F1	V F1	F F1
—	always predict N (patient)	0.236	0	n/a	0	0	0
E0	baseline (random split)	0.874 ± 0.009	0.618 ± 0.027	0.910 ± 0.017	0.736 ± 0.023	0.955 ± 0.005	0.814 ± 0.010
E1	baseline (patient)	0.429 ± 0.002	0.000 ± 0.000	n/a	0.000 ± 0.000	0.756 ± 0.008	0.000 ± 0.000
E2	+ weighted loss	0.371 ± 0.067	0.830 ± 0.060	0.115 ± 0.013	0.201 ± 0.018	0.629 ± 0.102	0.004 ± 0.002
E3	+ focal loss	0.444 ± 0.017	0.000 ± 0.000	n/a	0.000 ± 0.000	0.811 ± 0.067	0.001 ± 0.002
E4	+ augmentation	0.433 ± 0.017	0.000 ± 0.000	n/a	0.000 ± 0.000	0.771 ± 0.061	0.001 ± 0.003
E5	+ RR features	0.445 ± 0.030	0.001 ± 0.001	0.167*	0.001 ± 0.002	0.799 ± 0.092	0.027 ± 0.047

The same network that reaches macro-F1 0.874 on a random beat split reaches 0.429 on unseen patients (Table 2, Fig. 2). Accuracy hides this: it falls only from 0.980 to 0.923, 3.3 points above the 0.890 of always predicting N. As H1 predicted, the loss is concentrated in the rare classes. S F1 falls from 0.736 to 0 and F F1 from 0.814 to 0, while V F1 falls less, from 0.955 to 0.756. On unseen patients the baseline never predicts S: 98.6% of the 1,836 test S beats are called N, and the F beats of record 213 are split evenly between N and V.

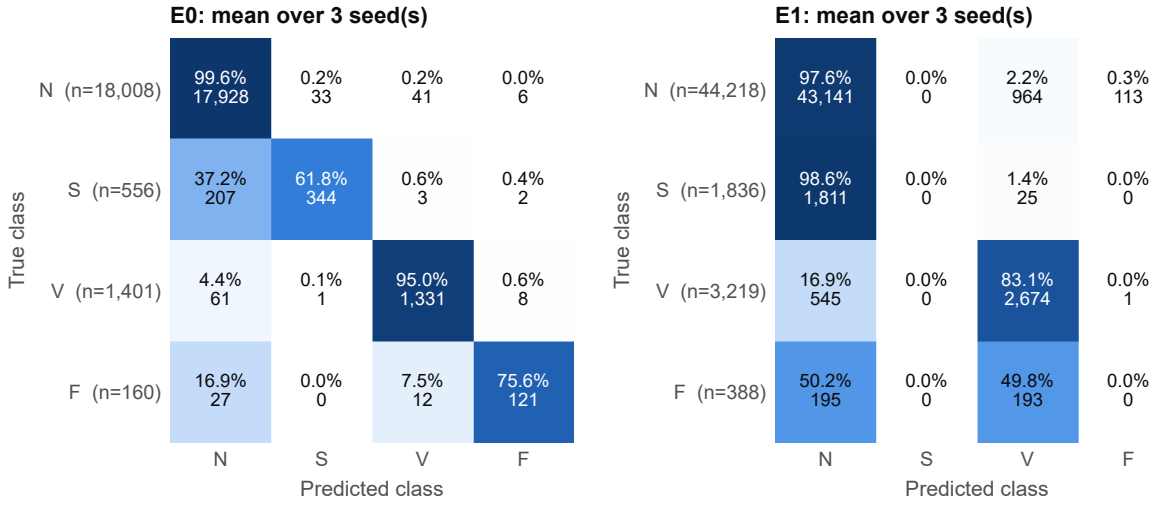


Figure 2: Confusion matrices of the same CNN on a random beat split (E0, left) and on unseen patients (E1, right), mean over three seeds. Cells show the percentage of the true class and the number of beats.

4.3 Effect of the fixes (H2–H5)

Table 3: Change from E1 with 95% patient-bootstrap intervals (2,000 resamples of the 22 test records, seed-averaged), and S and F F1 without their dominant records.

Exp	Δ Macro-F1	Δ S F1	Δ V F1	Δ F F1	S F1 without 232	F F1 without 213
E2	−0.058 [−0.135, +0.042]	+0.201 [+0.008, +0.473]	−0.127 [−0.227, +0.016]	+0.004 [0.000, +0.017]	0.027 ± 0.009	0.001 ± 0.001
E3	+0.015 [−0.001, +0.034]	0	+0.056 [−0.006, +0.126]	+0.001 [0.000, +0.020]	0	0.002 ± 0.003
E4	+0.004 [−0.003, +0.016]	0	+0.015 [−0.014, +0.049]	+0.001 [0.000, +0.051]	0	0.008 ± 0.013
E5	+0.016 [+0.004, +0.053]	+0.001 [0.000, +0.012]	+0.043 [+0.012, +0.079]	+0.027 [0.000, +0.172]	0.006 ± 0.010	0.001 ± 0.001

- **H2, partly supported.** Class weighting (E2) raises S sensitivity from 0 to 0.83, but S +P is 0.115, so about eight of every nine S predictions are false; macro-F1 falls by 0.058 (the interval includes zero), and V F1 and accuracy (0.530 ± 0.155) fall with it. Without record 232 the S gain almost disappears (S F1 0.027), and

Section 4.4 shows that even the gain in record 232 is not beat detection. Focal loss (E3) does not change S or F; its small gains are in V.

- **H3, supported.** Augmentation (E4) changes macro-F1 by +0.004 [−0.003, +0.016] and detects no S beats.
- **H4, not supported.** E5 predicts S for almost no test beat (S Se 0.001). Its macro-F1 gain, +0.016 [+0.004, +0.053], is the only macro-F1 interval that excludes zero, but it comes from V (+0.043), not from S.
- **H5, supported.** F F1 is at most 0.03 for every patient-split model; the largest, E5's 0.027, has an interval from 0 to 0.172 and comes from one seed. Training F beats come almost entirely from record 208 and test F beats from record 213.

Seed spreads are large for the patient-split models (for example ± 0.09 in V F1 for E5) and the bootstrap intervals are wide: which patients are tested matters as much as which fix is used. Fig. 3 shows S sensitivity for each test record.

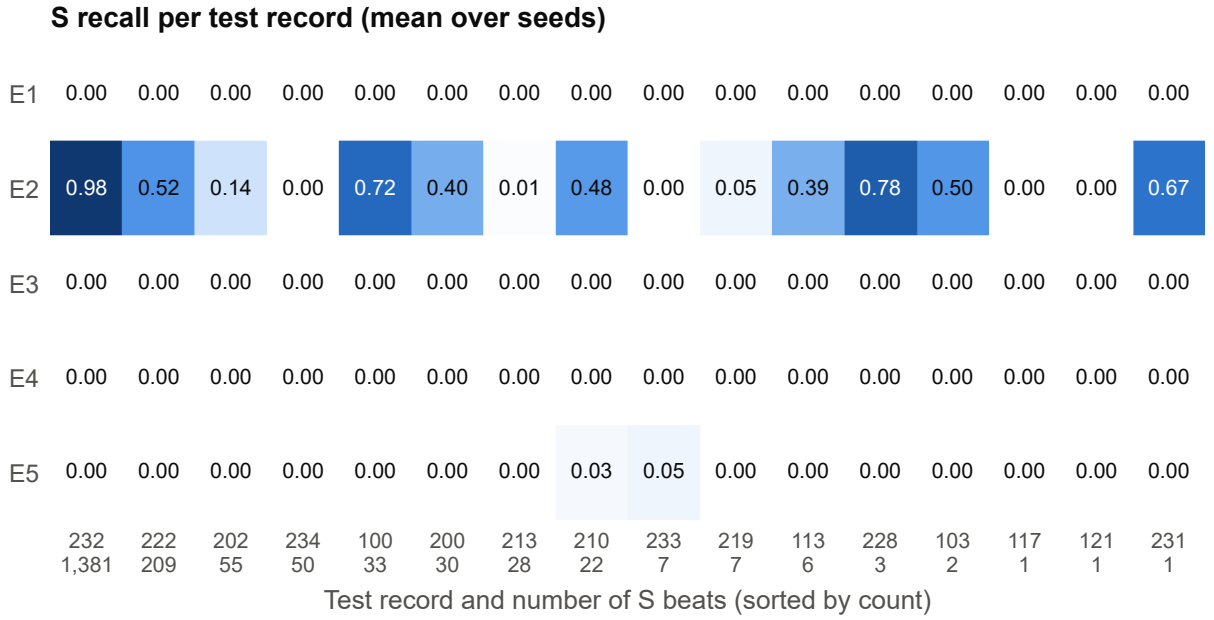


Figure 3: S sensitivity for each test record (mean over seeds), records sorted by their number of S beats (second line of the labels). E2's sensitivity in record 232 comes with S predictions for almost every beat of that record (Section 4.4).

4.4 Failure analysis

Where are the errors? For E1, two records account for 86% of the S errors: record 232 (1,379 S beats called N, 99.9% of its S beats) and record 222 (all 209 S beats called N). The other large error cells are record 213 (363 N beats called V; F beats split between N and V) and record 233 (202 V beats called N, 24%).

Did the models learn S on the training patients? On training and validation data only, the selected checkpoints of E1, E3, E4 and two of the three E5 seeds detect at most 3% of the S beats in their own training set. Early stopping kept epochs 1–5 in 14 of the 15 patient-split runs (epoch 9 in the other), against 18–29 for E0, because validation macro-F1 stopped improving almost at once while the validation loss of E1 rose from the first epochs (Fig. 4). The models that did fit S still fail on new patients: E2 detects 85–90% of its training S beats and E5 (seed 42) 61%, yet they find at most 8% of the S beats of the validation patients. Two failures therefore stack: model selection stops before S is learned, and S learned on the training patients does not transfer.

E2 labels patients, not beats *(post-hoc)*. Class weighting makes E2 call 98% of all beats in record 232 S, including 99% of that record's 397 N beats. Because 78% of the record's beats are S, this produces most of the S "gain". The same behaviour produces the false positives: 92% of the N beats of record 212 (2,530 of 2,746) are called S, as are 80% of those of record 222 and 68% of those of record 100. Within record 232, E2's S probability separates S from N beats no better than chance (AUROC 0.48 ± 0.06).

E5 ranks S beats correctly but never predicts them *(post-hoc)*. Table 4 measures how well each model's S probability ranks S beats above all other beats, across all test beats pooled and within each record. Pooled AUROC looks moderate for every patient-split model (0.75–0.86), but within records E1–E4 are at or below chance (median 0.46–0.48): their pooled score separates patients, not beats. E5 is the exception. Within record 232 its S probability separates S from N almost perfectly (AUROC 0.994), and the median over records is 0.895, yet the argmax never selects S.

Would a better threshold fix it? *(Post-hoc; thresholds chosen on validation patients only.)* We predicted S whenever the S probability exceeded the threshold that maximised S F1 on the four validation records. For no model does this transfer. For E1, E2 and E4 the threshold labels most test beats S (S Se 0.94–1.00, +P 0.04–

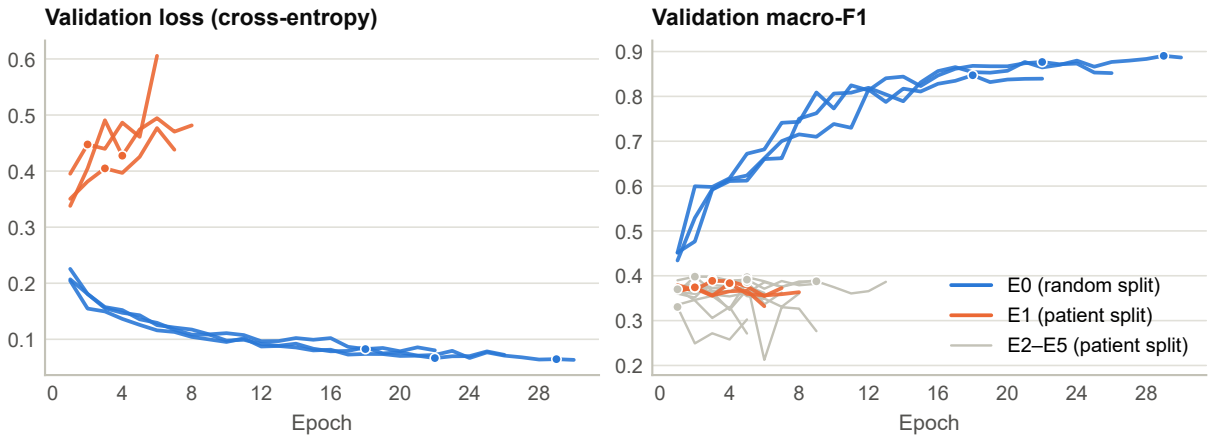


Figure 4: Validation loss (E0 and E1, cross-entropy) and validation macro-F1 (all experiments) per epoch; dots mark the epoch kept by early stopping. On the patient split, validation loss rises after the first epochs and macro-F1 never improves.

Table 4: *(Post-hoc.)* AUROC of the S probability for S versus all other beats, mean over seeds. Within records: median over the 8 test records with at least 10 S and 10 other beats.

Exp	Pooled	Within records (median)	Within record 232	Share of record 232 predicted S
E0 (random split)	0.963	0.943	0.948	0.775
E1	0.753	0.467	0.429	0.000
E2	0.859	0.478	0.482	0.983
E3	0.863	0.475	0.461	0.000
E4	0.762	0.457	0.419	0.000
E5	0.863	0.895	0.994	0.000

0.07). E3 reaches S F1 0.21 ± 0.11, but with within-record ranking at chance this again labels patients rather than beats. For E5 the outcome depends on the seed (S Se 0.62 ± 0.45, +P 0.11 ± 0.05, S F1 0.15 ± 0.08), and macro-F1 is no better than with the argmax (0.420 against 0.445). The scores are informative within a patient but not comparable between patients.

Why does timing not transfer? In the training set, 62% of S beats (395 of 640) occur inside supraventricular tachycardia runs, mostly in record 209, where they are not early relative to the preceding beats. Only 48.9% of training S beats have a prematurity index below 0.9, against 87.7% of test S beats (Fig. 5). Within record 232, S beats correlate at r = 0.975 with that record's N beats in shape; their timing is what differs. The cue that defines most test S beats is therefore weak in training.

5 Discussion and Limitations

The headline number depends on the split. The same 13.5k-parameter CNN, trained in the same way, scores macro-F1 0.874 when beats are split at random and 0.429 when the test patients are new, and on new patients it detects no S or F beats. This matches the direction reported in the literature [5], [6] and shows how far the random-split number overstates what a deployed model would do.

Why the inexpensive fixes did not recover S. Reweighting the loss (E2) did not teach the network what an S beat looks like in a new patient; it taught the network to call whole patients S. Imbalance is real, but it is not what blocks generalisation. Augmentation (E4) simulates gain, noise and baseline changes, while the S beats of record 232 differ from its N beats in timing, not shape, so, as expected, it did not help. RR features (E5) did teach the right cue: within record 232 the network ranks S above N almost perfectly. But no test beat crosses the decision boundary, and a threshold chosen on the validation patients does not help, because S scores are not comparable between patients. The training data teach a weaker timing cue than the test patients need.

Pooled metrics can mislead. The gap between pooled and within-patient AUROC (0.75–0.86 against 0.46–0.48 for E1–E4) shows that a pooled ranking metric can look reasonable when a model only separates patients with many S beats from patients with few. Within-patient analysis exposes this, and we recommend reporting it alongside pooled scores.

Model selection under patient shift. Early stopping on validation macro-F1 chose epochs 1–5 in 14 of 15 patient-split runs, before the network had learned S even on its training patients. This is itself a symptom of the shift: learning S on the training patients did not raise validation macro-F1, because the validation patients' S beats differ. Where S was learned (E2 and one E5 seed), validation S sensitivity stayed at or below 0.08, so the failure to generalise does not depend on the early stop. Still, E3–E5 partly compare under-trained models, and a selection rule that does not rely on four validation patients would make the comparison cleaner.

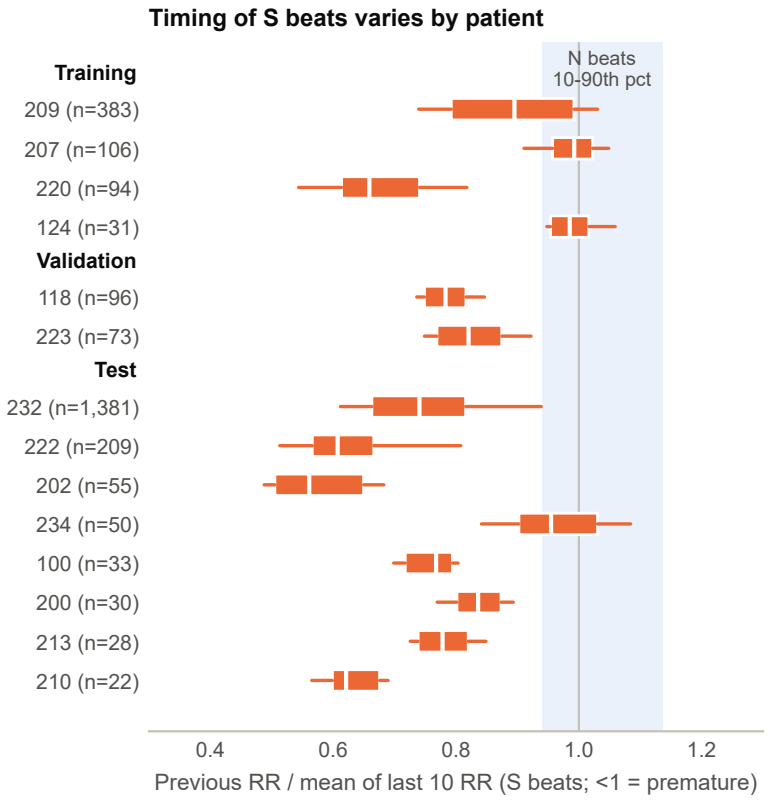


Figure 5: Prematurity index of S beats per record (previous RR divided by the mean of the last ten RR intervals; below 1 means early): median, interquartile range and 10th–90th percentiles. Training records 209 and 207 have S beats that are not early; most test S beats, led by record 232, are early.

Limitations. (i) *Few patients*: 22 test patients, two of whom hold most of the S (record 232: 75%) and F (record 213: 93%) beats, so S and F conclusions describe a handful of individuals, and a 22-record bootstrap is itself approximate [20]. (ii) *Weak validation set*: four patients with 175 S and 17 F beats guide early stopping and threshold choice. (iii) *Statistics*: three seeds per configuration, SDs up to 0.155 (E2 accuracy), intervals instead of significance tests, and no correction for comparing four fixes. (iv) *Data*: one lead, annotated beat locations instead of a detector, an offline next-RR feature, paced records and the Q class excluded, and no external test database. (v) *Post-hoc analyses*: the ranking and threshold analyses were designed after we saw the first test results; they explain the failure and were not used to select any model.

Future work. The within-patient result suggests that the missing step is per-patient calibration rather than more features: normalising RR features by each patient's own rhythm, adapting the threshold with a few labelled beats from the new patient, as in patient-specific training [21], or unsupervised test-time adaptation. Each needs a validation design with more patients, for example leave-one-patient-out on DS1.

6 Conclusion

On MIT-BIH, a 1D CNN that scores macro-F1 0.874 on a random beat split scores 0.429 on patients it has not seen and detects none of their S or F beats. Four inexpensive fixes did not change this. Class weighting raised S sensitivity to 0.83, but only by labelling entire patients S (S +P 0.115); focal loss and augmentation had no effect on S; RR features gave the only reliable macro-F1 gain (+0.016, through V). Their main effect was hidden: the model learned to rank S beats correctly within each patient (AUROC 0.994 in record 232) but never predicted them, and a threshold chosen on other patients did not fix this. Patient-wise evaluation should report per-patient results and within-patient ranking alongside pooled scores, because pooled scores can hide both failures and partial successes. The next step is to calibrate decisions per patient.

AI Use Statement

We used generative AI tools in all three roles, and each member checked the output of the tool they used.

M1 used OpenAI Codex for the data pipeline, autoencoder pretraining, documentation and report drafting. Outputs were verified using source-data checksums, protocol tests, encoder audits and checks against primary literature.

M2 used OpenAI Chatgpt 6 for environment set up, model creation, finding documentation, Readme and report drafting. Outputs were verified, checked, protocol tests and checks with primary literature.

M3 used Claude Code (Claude Opus 5.5) for the prediction-export, evaluation and figure code, the post-hoc analyses, reference checking and first drafts of Sections 1, 2.3, 3.5 and 4–6. The code was checked with unit tests (including cross-checks against scikit-learn), every reported number was regenerated from the code's output, and every reference was checked against the publisher, PubMed or arXiv record. [Yitian: how you reviewed and revised the text.]

Our team chose the research question, the subject-disjoint protocol and the experiments, decided how to report the early-stopping problem, and interpreted the results. Every result comes from experiments we ran.

References

[1] G. Moody and R. Mark, "The impact of the MIT-BIH Arrhythmia Database," *IEEE Engineering in Medicine and Biology Magazine*, vol. 20, no. 3, pp. 45–50, 2001. DOI: 10.1109/51.932724.

[2] A. L. Goldberger et al., "PhysioBank, PhysioToolkit, and PhysioNet: Components of a New Research Resource for Complex Physiologic Signals," *Circulation*, vol. 101, no. 23, e215–e220, 2000. DOI: 10.1161/01.cir.101.23.e215.

[3] Association for the Advancement of Medical Instrumentation, *ANSI/AAMI EC57:2012 Testing and Reporting Performance Results of Cardiac Rhythm and ST Segment Measurement Algorithms*, Arlington, VA, 2012.

[4] S. Mousavi and F. Afghah, "Inter- and Intra- Patient ECG Heartbeat Classification for Arrhythmia Detection: A Sequence to Sequence Deep Learning Approach," in *ICASSP 2019 - 2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2019, pp. 1308–1312. DOI: 10.1109/icassp.2019.8683140.

[5] P. de Chazal, M. O'Dwyer, and R. B. Reilly, "Automatic classification of heartbeats using ECG morphology and heartbeat interval features," *IEEE Transactions on Biomedical Engineering*, vol. 51, no. 7, pp. 1196–1206, 2004. DOI: 10.1109/TBME.2004.827359.

[6] A. Sellami and H. Hwang, "A robust deep convolutional neural network with batch-weighted loss for heartbeat classification," *Expert Systems with Applications*, vol. 122, pp. 75–84, 2019. DOI: 10.1016/j.eswa.2018.12.037.

[7] S. Saeb, L. Lonini, A. Jayaraman, D. C. Mohr, and K. P. Kording, "The need to approximate the use-case in clinical machine learning," *GigaScience*, vol. 6, no. 5, 2017. DOI: 10.1093/gigascience/gix019.

[8] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," *Patterns*, vol. 4, no. 9, p. 100 804, 2023. DOI: 10.1016/j.patter.2023.100804.

[9] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, and S. Belongie, "Class-Balanced Loss Based on Effective Number of Samples," in *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 9260–9269. DOI: 10.1109/cvpr.2019.00949.

[10] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollar, "Focal Loss for Dense Object Detection," in *2017 IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 2999–3007. DOI: 10.1109/iccv.2017.324.

[11] T. T. Um et al., "Data augmentation of wearable sensor data for Parkinson's disease monitoring using convolutional neural networks," in *Proceedings of the 19th ACM International Conference on Multimodal Interaction*, ser. ICMI '17, 2017, pp. 216–220. DOI: 10.1145/3136755.3136817.

[12] B. K. Iwana and S. Uchida, "An empirical survey of data augmentation for time series classification with neural networks," *PLOS ONE*, vol. 16, no. 7, e0254841, 2021. DOI: 10.1371/journal.pone.0254841.

[13] M. Buda, A. Maki, and M. A. Mazurowski, "A systematic study of the class imbalance problem in convolutional neural networks," *Neural Networks*, vol. 106, pp. 249–259, 2018. DOI: 10.1016/j.neunet.2018.07.011.

[14] U. R. Acharya et al., "A deep convolutional neural network model to classify heartbeats," *Computers in Biology and Medicine*, vol. 89, pp. 389–396, 2017. DOI: 10.1016/j.compbiomed.2017.08.022.

[15] N. Nonaka and J. Seita, *Data augmentation for electrocardiogram classification with deep neural network*, 2020. arXiv: 2009.04398 [eess.SP].

[16] A. Raghu, D. Shanmugam, E. Pomerantsev, J. Guttag, and C. M. Stultz, "Data augmentation for electrocardiograms," in *Proceedings of the Conference on Health, Inference, and Learning*, ser. Proceedings of Machine Learning Research, vol. 174, 2022, pp. 282–310.

[17] G. Moody and R. Mark, *MIT-BIH Arrhythmia Database*, version 1.0.0, 2005. DOI: 10.13026/C2F305.

[18] C. A. Field and A. H. Welsh, "Bootstrapping Clustered Data," *Journal of the Royal Statistical Society Series B: Statistical Methodology*, vol. 69, no. 3, pp. 369–390, 2007. DOI: 10.1111/j.1467-9868.2007.00593.x.

[19] A. C. Davison and D. V. Hinkley, *Bootstrap Methods and their Application*. Cambridge University Press, 1997. DOI: 10.1017/cbo9780511802843.

[20] B. Efron and R. Tibshirani, *An Introduction to the Bootstrap*. Chapman & Hall, 1993. DOI: 10.1201/9780429246593.

[21] S. Kiranyaz, T. Ince, and M. Gabbouj, "Real-Time Patient-Specific ECG Classification by 1-D Convolutional Neural Networks," *IEEE Transactions on Biomedical Engineering*, vol. 63, no. 3, pp. 664–675, 2016. DOI: 10.1109/tbme.2015.2468589.

A Reproducing the results

Data preparation (M1). From a fresh clone, run the following from the repository root; CPU is sufficient for preparation and audit:

```
python -m pip install -r requirements.txt
python -m src.prepare
python -m src.data audit
python -m unittest discover -s tests -v
```

The repository includes the 44 original raw recordings and reference metadata; large NPZ arrays are rebuilt locally. `src.prepare` verifies raw checksums, builds missing arrays with matching local metadata, and audits complete caches. Use `patient_strict_{train,val,test}.npz` for E1–E5 and `random_{train,val,test}.npz` for E0. Install a suitable PyTorch build to execute the model/E6 tests; they are skipped in a data-only environment.

NPZ byte hashes may differ across environments. Preserve any mismatched existing cache and rebuild into a new empty directory, for example `python -m src.prepare --output-dir data/processed_v2`. Pass the new directory to prediction with `--data-dir` and to evaluation with `--processed-dir`; do not commit locally regenerated reference metadata or edit hashes to bypass an audit failure.

Model training (M2). All six experiments (E0–E5) were trained using AdamW with a learning rate of 10^{-3}, weight decay of 10^{-4}, and batch size of 256. Training was limited to 30 epochs, with early stopping after four consecutive epochs without improvement in validation macro-F1. Each experiment was repeated with three seeds (42, 43, 44). E0 used the random split, while E1–E5 used the `patient_strict` protocol. Training requires an NVIDIA GPU with CUDA support in the current implementation. To reproduce the experiments, run from the repository root:

```
python -m src.train --experiments all --seeds 42 43 44 \
    --patient-protocol patient_strict --epochs 30 --patience 4 \
    --batch-size 256 --lr 0.001 --weight-decay 0.0001 --device cuda
```

The best checkpoint for each run is saved under `checkpoints/E*/seed*.pt`, with training histories and configurations stored in `results/E*/`.

The evaluation is reproduced from the repository root with:

- `python -m src.predict` exports test predictions from the committed checkpoints (CPU is enough).
- `python -m src.evaluate summarise --reference E1` writes every table and figure in this report to `results/summary/`.

B Autoencoder pretraining (E6)

M1 pretrained a convolutional autoencoder on training waveforms only, separately for the conventional and strict patient protocols with seeds 42, 43 and 44. Three stride-two convolutional stages with 16, 32 and 64 channels, GroupNorm and ReLU map each 256-sample beat to a 64×32 representation; a mirrored transposed-convolution decoder reconstructs the waveform. Training minimised MSE using AdamW (learning rate 0.001, weight decay 0.0001, batch size 256) for ten fixed epochs without augmentation. Labels, validation/test waveforms and validation-based epoch selection were excluded. The final-epoch encoders, complete histories and checksums are versioned in `artifacts/pretraining/`; all six exports were independently audited.

To audit the exports or repeat strict-protocol pretraining in a new output directory:

```
python -m pip install -r requirements-pretrain.txt
python -m src.pretrain_audit
python -m src.pretrain --protocol patient_strict --seeds 42 43 44 \
    --epochs 10 --output-dir local_pretraining
```

The continuation character above is for Bash/Colab; on other shells, place the training command on one line. A controlled E1/E6 comparison requires identical encoder/head architectures and matching training-data provenance, with only encoder initialisation changed. The exact archive-hash transfer check can reject arrays rebuilt in another environment. **No supervised E6 classifier was trained; H6 remains not tested.** Lower reconstruction MSE establishes optimisation of the pretraining objective, not improved classification or unseen-patient generalisation.

C Bootstrap differences

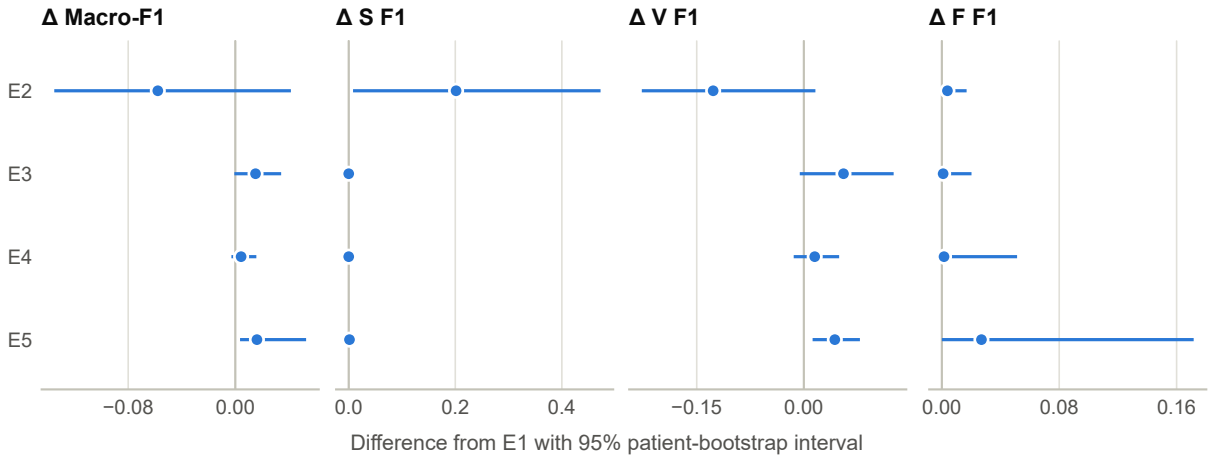


Figure C1: Change from E1 with 95% patient-bootstrap intervals (the data of Table 3).